Enterprise AI Digital Twin & Transformation Platform

Technical Architecture & System Documentation

Executive Overview

The **Enterprise AI Digital Twin Platform** is a multi-agent decision intelligence system designed to model, simulate, and optimize enterprise transformation strategies. By combining a 5-Agent LLM pipeline, a NetworkX graph traversal engine, and a Qdrant vector retrieval-augmented generation (RAG) system, the platform allows enterprise leaders to map value chains, evaluate AI automation potential, assess workforce skill transitions, and enforce multi-jurisdictional AI governance.

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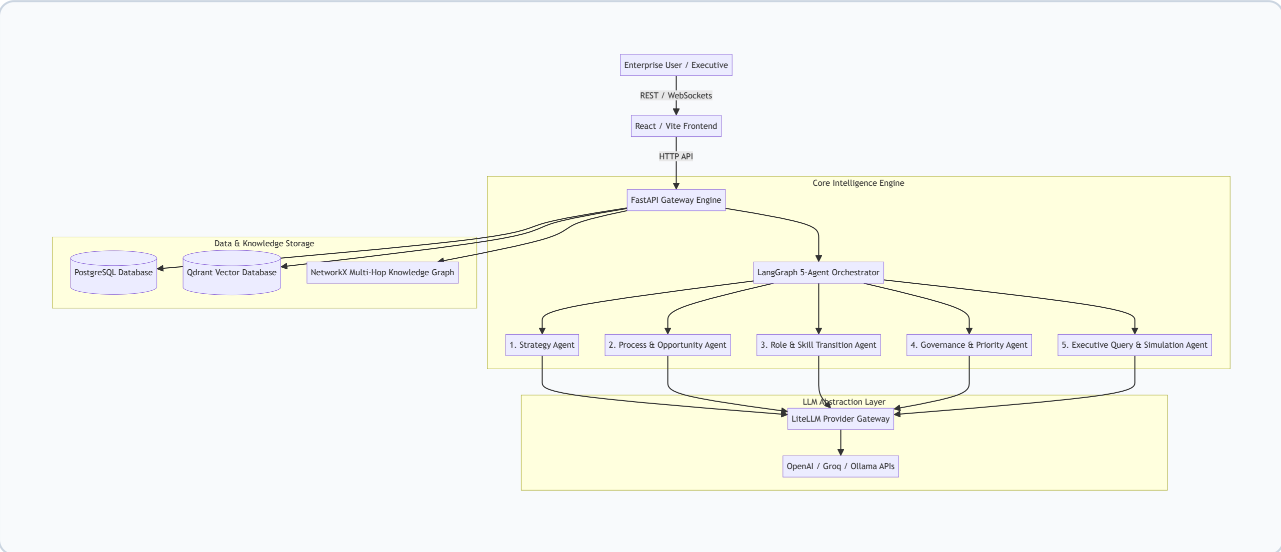
1. System Architecture



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2. 5-Agent Pipeline Topology

The core execution topology is orchestrated via **LangGraph**. When an enterprise strategy directive is submitted, data flows through five specialized agents sequentially:

Agent 1: Strategy Agent

- **Role**: Parses high-level strategic directives into structured operational objectives.
- **Outputs**: Value chain stage definitions, primary operational initiatives, target scope.

Agent 2: Process & Opportunity Agent

- **Role**: Decomposes value chain stages into core business processes and activity flows.
- **Outputs**: Sub-processes, activity sequences, automation potential scoring (`high`, `medium`, `low`), and concrete AI opportunity candidates.

Agent 3: Role & Skill Transition Agent

- **Role**: Analyzes organizational impact on human workforce and maps skill transformations across 6 taxonomy classes:
  1. `emerging` (New capabilities required)
  2. `increasing` (Growing demand)
  3. `ai_augmented` (Human + AI collaboration)
  4. `changing` (Core responsibilities shifting)
  5. `declining` (Automated routine tasks)
  6. `enduring_human` (Critical human-only judgment)

Agent 4: Governance & Priority Agent

- **Role**: Assesses legal, regulatory, and ethical risks against pre-seeded regulatory frameworks:
  - **EU AI Act** (Regulation 2024/1689 - Article 9 Risk Management & Article 14 Human Oversight)
  - **Digital Personal Data Protection Act 2023** (India DPDP Act Section 6 & 8)
  - **NIST AI Risk Management Framework** (AI RMF 1.0 GOVERN / MAP / MEASURE / MANAGE)
  - **OECD Principles on Artificial Intelligence** (Principle 1.2 Human-centred values)
  - **ISO/IEC 42001** (AI Management System)
- **Outputs**: Risk levels (`high`, `medium`, `low`), human signoff flags, priority scores ($0.0 - 1.0$), priority rationales.

Agent 5: Executive Query & Simulation Agent

- **Role**: Operates dynamically on user questions (`ask: [question]`) and scenario diff simulations.
- **Engine**: Performs 2-hop graph traversals over NetworkX, queries Qdrant vector evidence, and synthesizes executive advisory responses.

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3. API Endpoints & Core Specifications

Method	Endpoint	Description
`POST`	`/api/v1/ingest`	Triggers full 5-Agent pipeline on a strategic prompt.
`POST`	`/query`	Executes NetworkX graph traversal + Qdrant RAG query.
`POST`	`/simulate`	Runs hypothetical graph diff simulations (`automate_activity` , `delay_initiative`).
`GET`	`/api/v1/chats`	Retrieves historical chat sessions and execution thinking steps.
`POST`	`/api/v1/chats`	Persists session message history and graph state.

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4. Configuration & Deployment Setup

Environment Variables (`apps/api/.env`)

```
# Database Connections
DATABASE_URL="postgresql+psycopg2://postgres:postgres@localhost:5432/modus"
QDRANT_URL="http://localhost:6333"

# LLM Provider Configuration (Configured via LiteLLM)

# Option 1: OpenAI
LLM_PROVIDER="openai"
LLM_MODEL="gpt-4o"
OPENAI_API_KEY="sk-..."

# Option 2: Groq
# LLM_PROVIDER="groq"
# LLM_MODEL="groq/llama3-70b-8192"
# GROQ_API_KEY="gsk_..."

# Option 3: Ollama (Local Execution)
# LLM_PROVIDER="ollama"
# LLM_MODEL="ollama/llama3"
# OLLAMA_API_BASE="http://localhost:11434"
```

Running with Docker Compose

```
# Spin up PostgreSQL, Qdrant, FastAPI backend, and React web client
docker compose up -d
```

```
# Seed Digital Twin Benchmark Dataset (Meridian Retail Group)
python3 scripts/seed_pipeline.py
```